The Matter Action

The same square that produced gravity, applied to the whole carrier. Every hypercharge and every electric charge of one generation, from which directions can be traversed both ways.

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SECTION ONE

The square was never only about gravity

Paper 24 recorded that the framework's largest remaining gap was a matter action. It turns out to be the same gap as the one already filled, looked at from one step further back.

Movement Nineteen builds a defect from a propagating part weighted u and a held part weighted r, and squares it. Read on the geometry block that gives MacDowell-Mansouri gravity: a topological term, an Einstein-Hilbert term, and a cosmological constant term. Nobody asked what happens if the curvature in that defect is the whole carrier's curvature rather than the geometry block's.

D = u F + r E   D ^ D = u^2 (F^F) + 2ur (F^E) + r^2 (E^E)

Three terms, because a binomial squared has three terms. And Movement Twenty already sorts the carrier's operations into exactly three classes: what commutes with the decay is gauge, what connects distinct sectors is matter, what fails to commute is mass. Two threes, and they are the same three.

```
W_L = u^2  = 0.477457514063   Light        propagating, massless   ->  GAUGE
W_B = 2ur  = 0.427050983125   Boundary     sector-connecting       ->  MATTER
W_M = r^2  = 0.095491502813   Matter/held  cost of persistence     ->  MASS
```

The weights were named Light, Boundary and Matter in Movement Five, years of argument before Movement Twenty existed to say what the three sectors were. They were named for their sectors before anyone knew that is what they were naming.

The exact ratios follow with nothing to tune, because u and r are pinned by T4.

```
mass to gauge      W_M / W_L = (r/u)^2 = 1/5          exactly
matter to gauge    W_B / W_L = 2r/u     = 2/sqrt5      exactly
cosmological       Lambda = 3 W_B / 2 W_L = 3/sqrt5    exactly
```

SECTION TWO

What this fixes in MacDowell-Mansouri

The MacDowell-Mansouri construction has a known weakness and it is not a small one. To get Einstein-Hilbert out of the square rather than a topological term, the action must be projected: the larger symmetry is broken down to the Lorentz subalgebra by an operator inserted by hand. Without that projection the whole thing is a total derivative and describes nothing. Critics have pointed at that insertion for fifty years and they are right to.

In this framework the projection is not inserted. It is the cluster projection, and Paper 21 forces which one it is: averaging over the Weyl group of the decayed subsystem. The symmetry breaking MacDowell-Mansouri assumes is the descent this corpus derives.

the decay is the projection MacDowell-Mansouri had to assume

That is the paper's strongest structural claim and it costs nothing extra: the projection was already there, doing a different job under a different name, for reasons that had nothing to do with gravity.

> *PROPOSED as the identification of the two projections; PROVED that the framework's projection exists, is unique given the decayed subsystem, and has the required fixed space. Certified by scripts/01_closure.py and scripts/matter_action.py.*

SECTION THREE

Hypercharge, forced

Paper 22 split the surviving five directions three against two on the criterion of reversibility: three spatial modes traversable both ways, and the clock and the closure, which are not. The gauge stabiliser of that split is the Standard Model group. Now ask what a hypercharge can be on such a split.

It must be constant on each block, since it cannot distinguish directions the decay does not distinguish, and it must be traceless to sit inside the unimodular algebra. Two conditions, two unknowns, and the ratio is fixed.

3a + 2b = 0 so a / b = -2/3

With the ordinary weak normalisation of one half, that is minus one third on each colour direction. The ratio is the content and it is forced; the overall normalisation is the usual convention and is not derived, and the paper says so rather than claiming a whole number it has only half earned.

SECTION FOUR

One generation, every charge

One generation is the even exterior algebra of the five complex directions, and a wedge carries the sum of the hypercharges of the directions in it. Electric charge is the third weak component plus hypercharge. Nothing else is put in. What comes out was checked against the Standard Model written out independently.

```
state   SU(3)  SU(2)     Y  dim  Q values      basis
Q           3      2   1/6    6  -1/3, 2/3     c1 ^ w1
u^c      3bar      1  -2/3    3  -2/3          c1 ^ c2
e^c         1      1     1    1  1             w1 ^ w2
d^c      3bar      1   1/3    3  1/3           c1 ^ c2 ^ w1 ^ w2
```

```
L           1      2    -1/2    2    -1, 0         c1 ^ c2 ^ c3 ^ w1
nu^c        1      1      0     1    0             1
                         total 16

all sixteen states match the Standard Model: True
```

The down quark's charge of one third is three colour directions and two weak directions wedged together. The electron's charge of one is the two weak directions alone. The neutrino's zero is the empty wedge, the state with no directions in it at all, which is why the right-handed neutrino is present rather than appended: it is what you get by taking nothing.

Quark charges are thirds because there are three colour directions and hypercharge has to be traceless across three of them and two of the others. That is the whole explanation. Fractional charge stops being a curiosity and becomes division.

SECTION FIVE

The weak mixing angle

The same split fixes the hypercharge normalisation and therefore the mixing angle at unification.

```
Tr(Y^2) over the 3+2 block = 3(-1/3)^2 + 2(1/2)^2 = 5/6
Tr(T3^2)                                         = 1/2
normalisation Tr(Y^2)/Tr(T3^2)                   = 5/3
sin^2(theta_W) at unification                    = 3/8
```

Three eighths is the standard grand-unified value and the framework reproduces it, which is what it should do. What is different is where the three and the two came from. In an ordinary unified model the split into colour and weak is a choice of breaking direction. Here it is the difference between directions that can be traversed both ways and directions that cannot, and that difference was fixed by an argument about records long before any of this.

One observation offered at its own low grade: three eighths is also the number of spatial modes over the rank of the carrier. Whether that is structure or arithmetic is not settled here and the corpus does not promote a resemblance to a derivation.

SECTION SIX

CP, and a term that has to be zero

The u squared term is the square of the curvature, and for a gauge sector that is the topological term whose coefficient is the vacuum angle. The coefficient is universal in this action, so the framework has no freedom to make any particular vacuum angle small. It has to make one of them zero for a reason, or fail.

Paper 22's sorting supplies the reason. A term whose entire content is sensitivity to orientation needs an orientation to be sensitive to. The strong sector lives on the three freely reversible directions and has none.

```
sector    block                        orientation available    observed
strong    three reversible modes       no                       theta < 1e-10
weak      clock and closure            yes                      CP violated
EM        traceless across both        no net                   CP conserved
gravity   the metric block             no net                   CP conserved
```

Two for two on the sectors where it matters, and no freedom was used in either. The strong vacuum angle is measured below one part in ten billion and has no accepted explanation; an entire proposed particle exists to explain it. On this reading it is not small, it is absent, and its absence is the same fact as parity being conserved there. And the weak sector, which lives on the arrow, is required to violate CP rather than permitted to.

> *PROPOSED, and the most valuable claim in the paper for exactly that reason. Deriving the vanishing of the strong vacuum angle from the reversibility of the colour block, rather than observing that the two are compatible, is a calculation this paper does not perform. It is now the sharpest open target in the corpus.*

SECTION SEVEN

What is new here and what is borrowed

The corpus is careful about this elsewhere and will be careful here, because the temptation to blur it is strongest when the result is prettiest.

Borrowed The exterior algebra description of one generation, the sixteen of so(10), the three eighths, the hypercharge pattern, and the MacDowell-Mansouri form itself. All of it is textbook and all of it has been written down by people who were not doing this.

New That the three-two split is derived from reversibility rather than chosen as a breaking direction. That the three weights of Movement Five are the three sectors of Movement Twenty, with exact ratios of one fifth and two over root five. That the projection MacDowell-Mansouri inserts by hand is the cluster projection. And that the strong vacuum angle vanishes structurally rather than by tuning.

The novel content is the forcing, not the group theory. The group theory then does what it has always done, which is the strongest position to be in, because it means the new step lands on ground that other people have already mapped and can check without taking anything on trust.

SECTION EIGHT

What the action still does not do

No field equations The action is identified and its coefficients are fixed. Nobody has varied it. Until someone does, this is a Lagrangian in the sense of an expression rather than in the sense of a dynamics.

No coupling The overall normalisation still sets the fine structure constant and the Rydberg and Newton's constant, and it is still one debt owed once rather than three owed separately. Paper 24 said that and this paper does not improve it.

No masses The weight of the mass sector is r squared and the ratio to the gauge sector is exactly one fifth. That is a statement about the action's coefficients, not about any particle's mass. The Yukawa structure and the hierarchy remain untouched.

No derivation of the vanishing vacuum angle Section Six identifies why it should vanish. It does not compute that it does.

So the largest gap named in Paper 24 is narrowed rather than closed: the framework now has one expression covering gravity, gauge, matter and mass, with every relative coefficient fixed and nothing tunable, and it does not yet have equations of motion. That is a better place to be stuck than the previous one, and it is the last place this sequence of papers reaches.